Artificial Intelligence and Data Science

Data Science and Visualization

24ADI204

DATA-VIZ HACK-A-BIT (2025-2026)

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Project Walkthrough

01. Project Scope

02.Problem Statement

03. Objectives

04. Project Workflow

05. Dataset Overview

06. Target Variable

07. Data Cleaning

08. Outlier Handling

09. EDA

10. Correlation Analysis

11. Feature Engineering

12. PCA

13.Key Insights /Results

14.Dashboard

15.Demo Video

16.Future Enhancements

17.Team Contribution

18.Conclusion/Reference

Project Scope

- This project analyzes 120 years of Olympic athlete data to uncover patterns in performance, participation, and medal outcomes.
- It aims to transform historical data into actionable insights that support data-driven decision-making in sports.
- The project focuses on identifying peak performance age, sport-specific physical attributes, and participation trends to improve athlete selection and training strategies.
- It also helps optimize funding decisions by highlighting high-probability medal scenarios, thereby reducing investment risk.
- Additionally, the project provides insights into evolving trends such as gender participation growth, making it relevant for both strategic planning and future sports development.
- It enables the use of data science techniques such as EDA, correlation analysis, and feature engineering to extract meaningful insights from large-scale datasets.

Problem Statement

- Over 120 years of Olympic data shows trends in athlete performance, geopolitical dominance, and physical evolution. However, organizations lack a structured, data-driven approach to
- Identify high medal probability athletes
- Allocate funding efficiently
- Optimize sport-specific investments
- Out of 271,116 records, only ~14.7% resulted in medals, highlighting the need for predictive and analytical frameworks
- "**The project addresses the problem of inefficient and intuition-based decision-making in sports, where organizations struggle to identify high medal-probability athletes and allocate funding effectively. As seen in the case of UK Sport after the 1996 Olympics, the absence of data-driven strategies can lead to poor performance, while adopting analytical approaches significantly improves medal outcomes. This highlights the need for transforming historical Olympic data into actionable insights for better strategic planning**."

Objectives

Objectives:

- Analyze historical Olympic data to uncover key performance patterns and trends
- Examine the relationship between age, physical attributes, and medal outcomes
- Identify the optimal performance age range for elite athletes
- Evaluate gender participation trends and their evolution over time
- Generate data-driven insights to support strategic decision-making and investment in sports

Project Workflow

Project Workflow

Data Collection (Kaggle dataset)

Data Cleaning (missing values, duplicates, inconsistencies)

Outlier Handling (IQR & Z-score methods)

Exploratory Data Analysis (EDA)

Feature Engineering (BMI creation)

Correlation Analysis

Visualization & Insight Generation

Dataset Overview

Dataset Overview
Dataset: *120 Years of Olympic History*
Source: Kaggle
Size: 271,116 rows, 15 columns and Data merged using NOC key
Features include:
Numerical: Age, Height, Weight, Year
Categorical: Sport, Event, Medal, Sex, Region
Key Features :
Numerical Features:
Age, Height, Weight, Year
Categorical Features:
Sport, Event, Medal, Sex, Region

Target Variable

The target variable is **Podium_Finish**, categorized as:
Medal
No Medal
This helps classify athlete success and analyze performance probability.

Data Cleaning

7. Data Cleaning

Data Preparation Steps

- Duplicate Handling

Removed 1,480 duplicate records to ensure data consistency

- Missing Value Treatment

Age → Median Imputation *(skewed distribution)*
Height → Mean Imputation *(symmetric distribution)*
Weight → Median Imputation

- Data Standardization

Cleaned and standardized region names for consistency

- Feature Selection

Dropped *'notes'* column due to high missing values

Outlier Handling

Outlier Handling

Outlier Detection & Treatment

1. **Age (Domain-Aware Handling)**

Detected outliers using **IQR method**
Applied **winsorization (clipping)** based on real-world limits
Final range: **10 – 73 years** *(historically valid athletes retained)*

2. **Height (Statistical Approach)**

Identified outliers using **Z-score and IQR methods**
Applied **capping using statistical bounds**
Ensured realistic distribution without extreme distortion

3. **Weight (Robust Handling)**

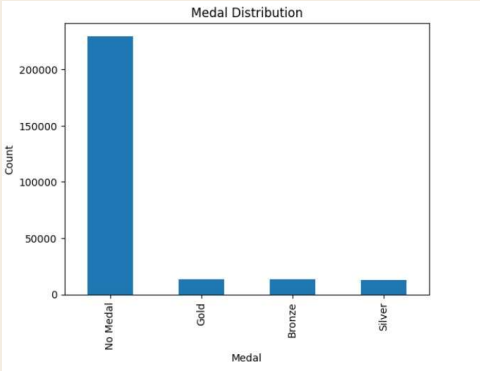
Detected outliers using **IQR method**
Applied **clipping to reduce skewness**
Preserved all records while limiting extreme values

Outcome

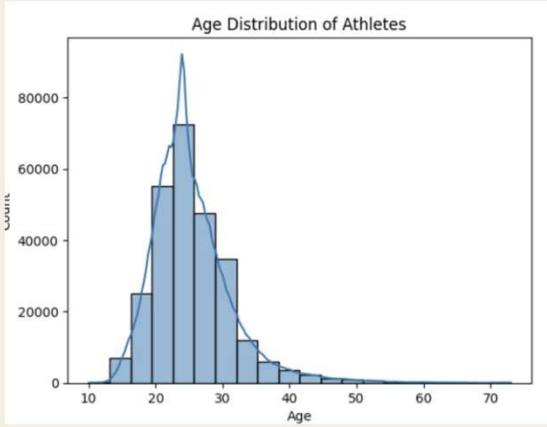
✓ Removed extreme distortions
✓ Preserved real-world valid data
✓ Improved data quality for accurate analysis

Exploratory Data Analysis

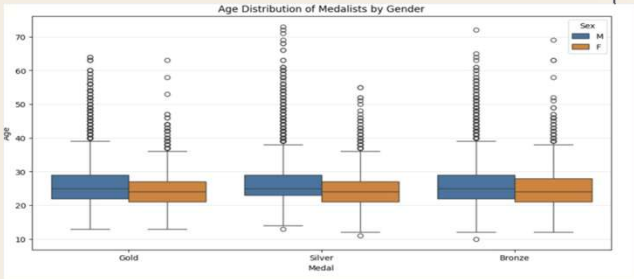
1. Bar Chart (Medal Distribution)
- **Univariate EDA**
One variable → Medal
Shows imbalance (No Medal >> others)



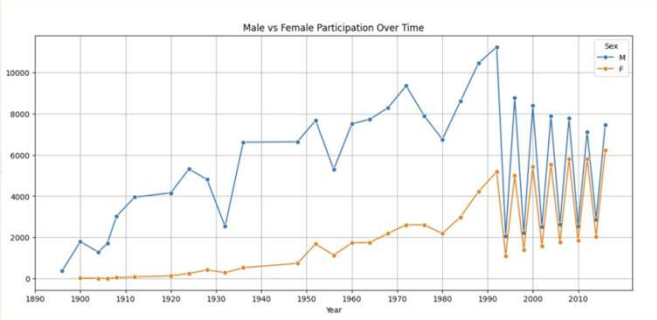
2. Histogram (Age Distribution)
Univariate EDA
Distribution + skewness
Mean vs Median insight



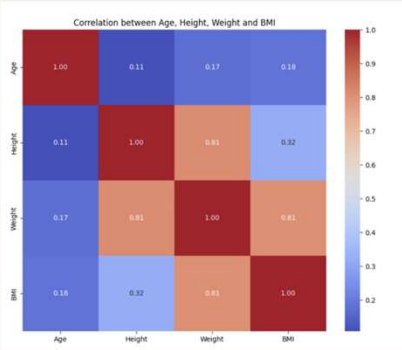
Exploratory Data Analysis



3. Boxplot (Gender vs Age/BMI etc.)
Bivariate EDA
Comparing **2 variables**
Shows spread + outliers



4. **Time-Series Analysis**
Analyzed trends over years using **line plots**
Observed increase in participation over time
Identified growth in **female participation**



5. **Multivariate Analysis**
Used **correlation heatmap** to analyze relationships between variables Found strong correlation between **Height and Weight (≈ 0.8)** Weak correlation between **Age and body metrics**

Exploratory Data Analysis

After analyzing 120 years of Olympic athlete data, we can summarize our findings into five key insights:

1. Peak Performance Window is Predictable- Majority of medal winners fall between ages 24–27- Performance probability declines beyond early 30s- Age shows weak correlation with height (0.11) and weight (0.17)Elite performance is not random — it follows a biological optimization window
.Talent investment should focus on athletes entering this age band.
2. Physical Structure Matters More Than Age- Strong correlation between height and weight: r = 0.81- Body metrics are highly interdependent- Age has minimal structural influenceAthlete selection and training models should prioritize physical profiling over age-based assumptions.This supports data-driven scouting systems
.3. Sport-Specific Body Types Exist- Basketball → Taller athletes dominate- Gymnastics → Shorter athletes dominate- Swimming/Athletics → Moderate buildThere is no universal "ideal athlete."Each sport has a biomechanical signature.
This allows:- Precision recruitment- Optimized talent identification- Reduced training investment risk

4. Gender Participation Has Transformed- Early Olympics: <5% female participation- Modern Olympics: ~45–48% representationBusiness Meaning:Women's sports are a high-growth investment segment
.Organizations investing in female sports programs now are entering an expanding market with long-term returns.

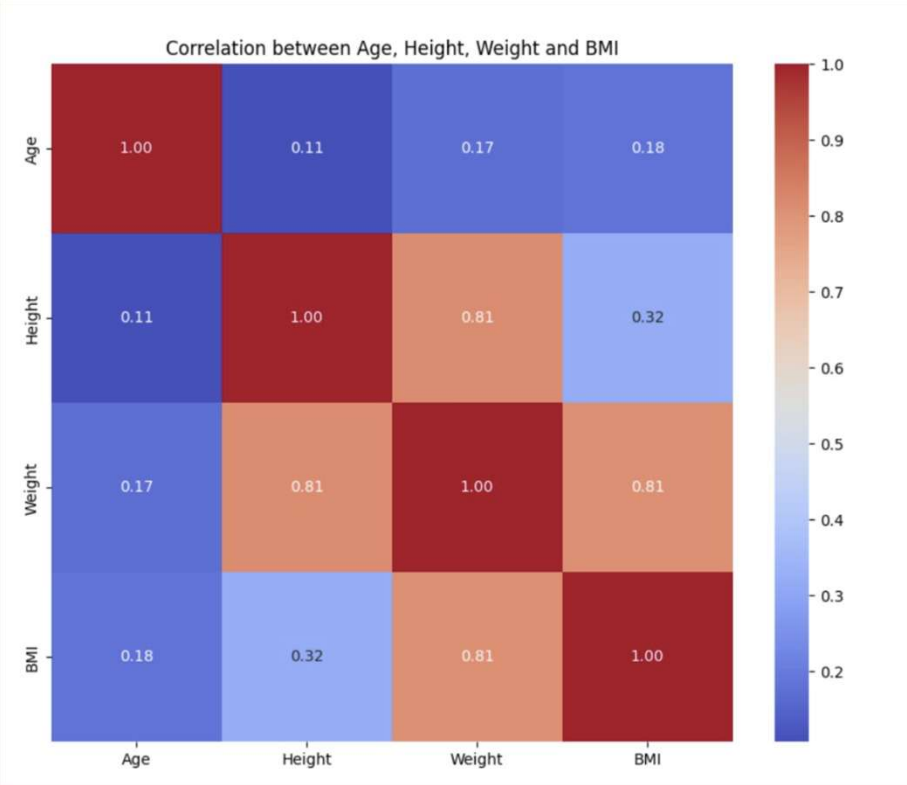
5. Medal Distribution Reflects Structural AdvantageDominance is not accidental — it reflects:- Economic infrastructure- Long-term training ecosystems- National investment strategyPerformance success correlates with sustained funding models
.This validates strategic, long-term athlete development programs
.Final Strategic ConclusionFrom a data science and business standpoint
:✓ Elite performance is predictable
✓ Athlete success can be modeled
✓ Gender equity is a growth opportunity
✓ Investment in physical profiling increases medal probability

Correlation Analysis

Objective
To analyze the relationship between numerical variables and understand how they influence each other.
Method Used
Pearson Correlation Coefficient
Visualized using a **correlation heatmap**

Key Observations
Height vs Weight
Strong positive correlation (**r ≈ 0.8**)
Taller athletes tend to weigh more
Age vs Height
Very weak correlation (**r ≈ 0.11**)
Age does not significantly influence height
Age vs Weight
Weak positive correlation (**r ≈ 0.17**)
Minimal impact of age on weight

Conclusion
Physical attributes are strongly related to each other
Age is relatively independent of body structure
Body metrics are more important for athlete profiling



Feature Engineering

Objective

To enhance data representation by creating meaningful features and improving the analytical quality of the dataset.

Feature Identification

•Classified variables into:

- **Numerical Features** → Age, Height, Weight, BMI, Year
- **Categorical Features** → Sport, Event, Medal, Sex, Region

Feature Creation

Body Mass Index (BMI)

•Derived using Height and Weight

•Represents athlete **body composition** more effectively

Statistical Analysis

•Performed **skewness analysis** on numerical features

•Identified **Age** as highly skewed

•Other features showed near-normal distribution

Feature Transformation Strategy

•**Age** → Log transformation (to reduce skewness)

•**Year** → Binning into decades (for better interpretability)

•**Height, Weight, BMI** → Retained (stable distribution)

Outcome

- Improved feature representation
- Enhanced interpretability of data
- Prepared dataset for advanced analysis and modeling

Dimensionality Reduction

Dimensionality Reduction
Objective:
To reduce the number of features while preserving important information, thereby improving efficiency and model performance.

Technique Studied: Principal Component Analysis (PCA)
PCA transforms original features into a smaller set of **uncorrelated components**
Helps in reducing **redundancy and multicollinearity**
Retains maximum variance in the data

Application in This Project
PCA was **explored conceptually** to understand feature relationships
Not fully applied, as the dataset had:
- Limited number of numerical features
- High interpretability requirement

Justification
Original features were already **meaningful and interpretable**
Applying PCA would reduce **explainability of insights**
Focus remained on **data analysis and visualization**

Key Insights / Results

Key Insights & Results

•Olympic performance shows a clear pattern, with most medal winners falling in the **22–27 age range**, indicating a peak performance window

•**Physical attributes play a major role**, with a strong correlation between height and weight, while age has minimal influence

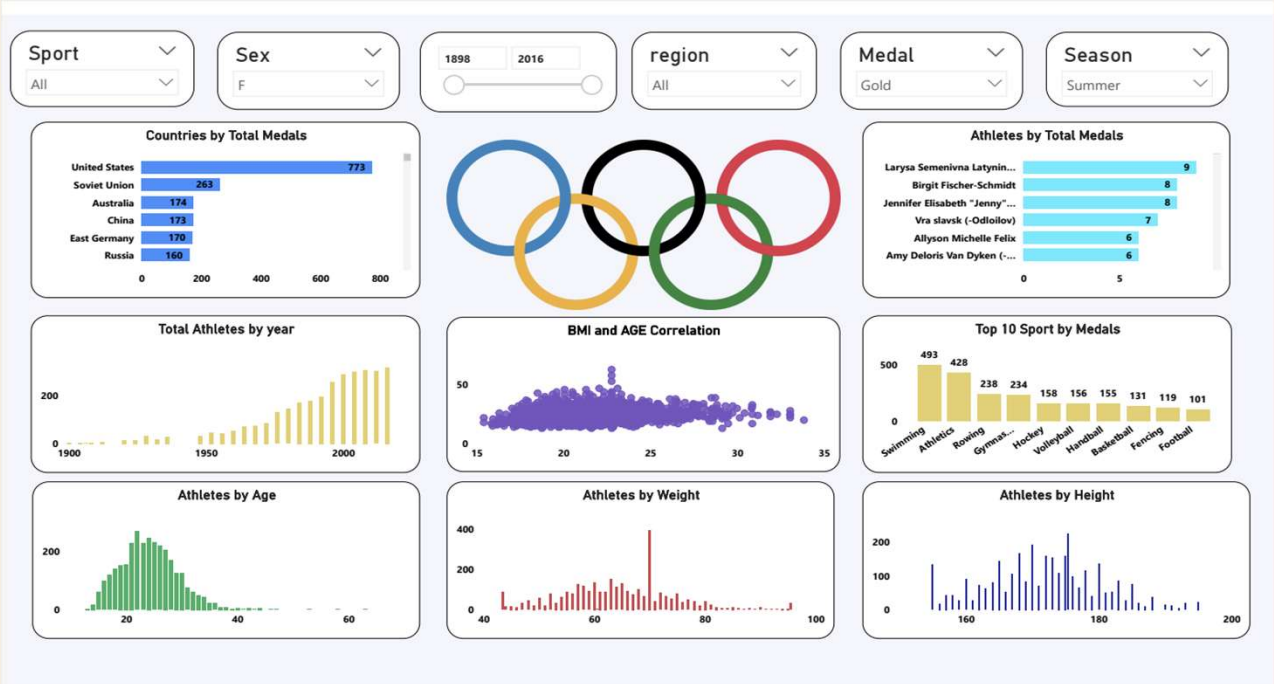
•Different sports require **specific body types**, proving there is no single ideal athlete profile

•**Medal distribution is highly imbalanced**, with the majority of athletes not winning medals, highlighting the importance of strategic selection

•**Country dominance is evident**, with nations like the United States consistently leading due to strong infrastructure and investment

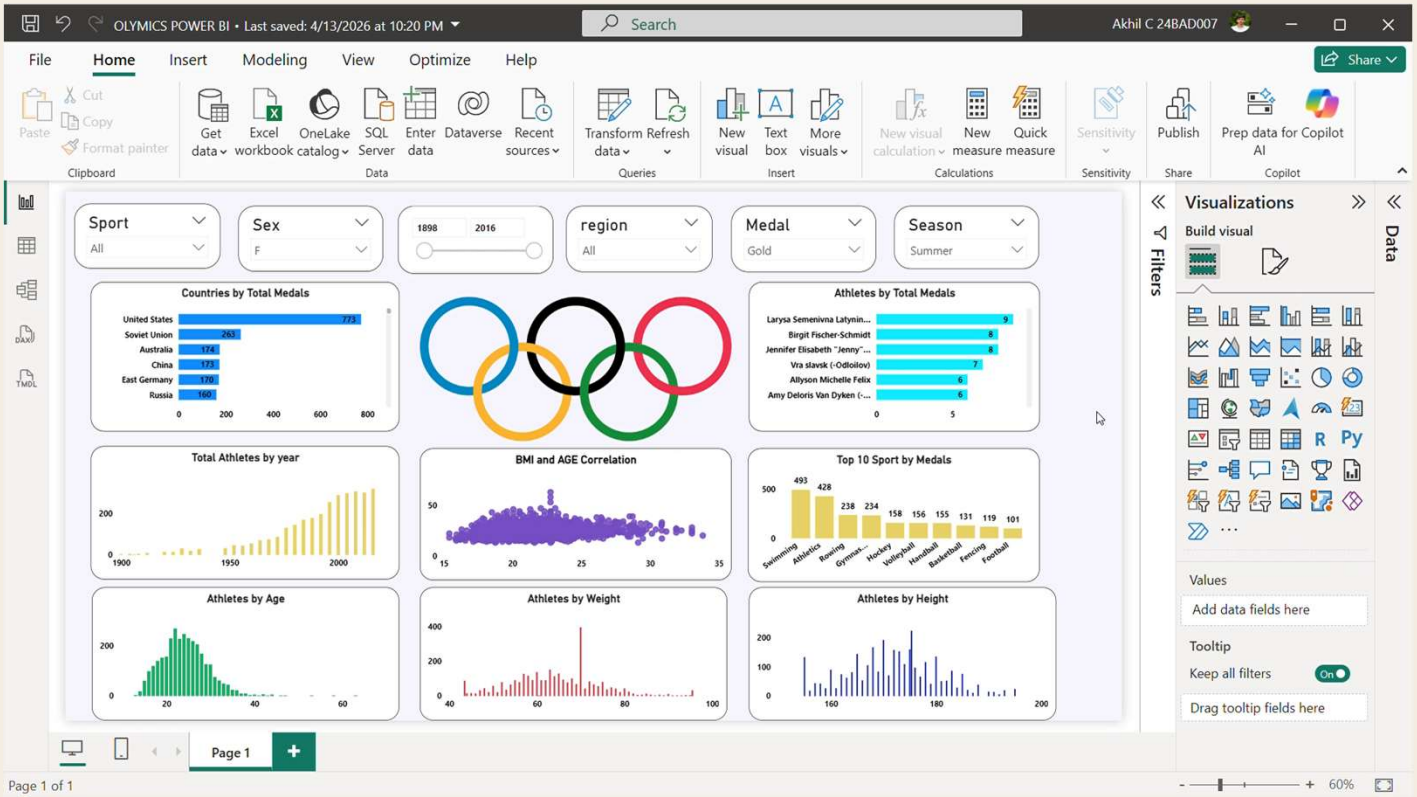
•**Female participation has increased significantly**, reflecting growing inclusivity and expansion of global sports

Visualization Dashboard



The analysis reveals that Olympic performance is dominated by a few countries like the USA, supported by strong infrastructure and investment. Athlete participation has significantly increased over time, reflecting global expansion of the Olympics. Most athletes fall within the 20–30 age range, indicating a clear peak performance window, while physical attributes like BMI show no strong relationship with age. Overall, performance is influenced more by sport-specific factors and structured development systems than by age alone.

Demo Video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Soviet Union
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China
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Russia
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Total Athletes by year
BMI and AGE Correlation
Top 10 Sport by Medals
Athletes by Age
Athletes by Weight
Athletes by Height
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Page 1 of 1
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Future Enhancements

Future Enhancements:

- Develop machine learning models to predict medal probability and athlete performance
- Build an interactive dashboard (Power BI/Tableau) for real-time exploration and decision-making
- Integrate real-time and recent Olympic data to improve relevance and accuracy
- Include additional factors such as training data, diet, injury history, and performance metrics
- Perform country-wise and sport-wise predictive analysis for strategic investment planning

Individual Contribution

Akhil C (24BAD007)
Performed data collection and dataset understanding
Handled data cleaning and preprocessing steps
Worked on feature engineering (BMI creation)
Assisted with reports

Kaviya S (24BAD059)
Conducted exploratory data analysis (EDA) and visualizations
Derived insights and interpretations from data
Prepared project presentation

Hamphiritha BS (24BAD403)
Assisted in outlier detection and treatment
Performed correlation analysis and interpretation
Contributed to dashboard creation and final outputs
Asssisted with reports

Conclusion and References

Conclusion

This project demonstrates that Olympic performance is not random but follows identifiable patterns based on age, physical attributes, and participation trends. Through data preprocessing, exploratory analysis, and feature engineering, meaningful insights were derived to support data-driven decision-making in sports. The analysis highlights that peak performance occurs within a specific age range, physical characteristics vary across sports, and medal success is highly competitive. Overall, the project emphasizes the importance of structured data analysis in improving athlete selection, optimizing resource allocation, and enabling strategic sports planning.

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•Course Materials – *Data Science and Visualization*
•*Lecture Slides (Faculty Provided)*